

AAPOR Report on Online Panels

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3. AN OVERVIEW OF ONLINE PANELS

One of the first challenges the researcher encounters in all survey modes is the development of a sample frame for the population of interest. In the case of online, survey researchers face the additional challenge of ensuring that sample members have access to the mode of questionnaire administration, that is, the Internet. For some potential populations of interest, almost all members are online (e.g., software developers, college students, and business executives). But if the research requires a general population sample of the US, not everyone is online. Estimates of Internet use and penetration in the U.S. household population can vary widely. Arguably the most accurate are those collected face-to-face by the Current Population Survey (CPS). The CPS reports that as of October, 2009, 69 percent of U.S. households had an Internet connection while 77 percent had household members who reported that they connected to the Internet from home or some other location such as their workplace (Current Population Survey, 2009). This comports with the most recent data from the Pew Research Center showing that, as of December of 2009, 74 percent of U.S. adults use the Internet either at home or some other location (Rainie, 2010). However, the Pew data also report that only 72 percent of Internet users actually go online at least once a week. Further, Internet access tends to be positively associated with income and education and negatively associated with age (younger people are more likely to be online than older people). Some demographic groups are also less likely to be online (e.g., blacks, Hispanics, and undocumented immigrants).

There also is no comprehensive list of those who are online. While virtually all Internet users have an email address, no complete list of these addresses exists. Individuals may have several email addresses or they may share a single address with other family members. Addresses may fall into disuse without being deactivated. In addition, the non-standardized format of e-mail addresses precludes the generation of RDD-like methods for Web surveys.

Even if such a comprehensive list existed there are both legal prohibitions and industry practices that discourage the kind of mass emailing that might be akin to the calling we do for an RDD telephone survey. The CAN-SPAM Act of 2003 established clear guidelines in the U. S. around the use of email addresses in terms of format, content, and process. Internet Service Providers (ISPs) generally take a dim view of mass emails and will sometimes block suspected spammers from sending email to their subscribers. The Council of American Survey Research Organizations (CASRO) has established the standard for its members that all email communications must be “permission-based.” Specifically, the CASRO Code of Standards and Ethics for Survey Research (2009) requires that its members only mail to potential respondents with whom the research organization or its client has a pre-existing relationship. Examples include individuals who have previously agreed to receive email communications either from the client or the research organization or customers of the client.

There are many specialized populations for which a full list of email addresses might be available and usable. Some examples include members of an organization (e.g., employees of a company or students at a university), users of a particular Web site, or customers of an online merchant. These circumstances are now relatively common. However, gaining access to representative samples of the general population for online research continues to be problematic.

Online panels have become a popular solution to the sample frame problem for those instances in which there is no usable and complete list of email addresses for the target population. For purposes of

this report we use the definition of online panel from ISO 26362: Access Panels in Market, Opinion, and Social Research. It reads: “A sample database of potential respondents who declare that they will cooperate with future [online] data collection if selected” (International Organization for Standardization 2009). As noted in Section 2 of this report, we further distinguish between *nonprobability-based panels* and *probability-based panels*. The former are panels constructed through the widespread placement of offers to join for the purpose of participating in future surveys. Anyone seeing the panel offering may choose to join and become a panel member provided they meet requirements specified by the panel builder. Probability-based panels select potential members in advance from a sampling frame of the target population and then attempt to recruit only those sampled individuals to join the panel and participate in future surveys, sometimes providing Internet access to those sampled members without access at the time they are contacted.

Probably the most familiar type of online panel is a *general population panel*. These panels typically include hundreds of thousands to several million members and are used for both general population studies as well as for reaching respondents with low incidence events or characteristics (e.g., owners of luxury vehicles or people suffering from Stage II pancreatic cancer). The panel serves as a frame from which samples are drawn to meet the specific needs of particular studies. The design of these study-specific samples may vary depending on the survey topic and population of interest.

Census-balanced samples are designed to reflect the basic demographics of the U.S. population (and the target proportions could be based on distributions as they occur in the larger population for some combination of sex, age, region of the country, income, education, ethnicity, or other relevant demographic characteristics). Another common example of an online panel is a *specialty panel*. A specialty panel could be a group of people who are selected because they own certain products (e.g., big screen TVs), are a specific demographic group (e.g., Spanish-speaking Hispanics in the U.S.), are in a specific profession (e.g., physicians), engage in certain behaviors (e.g., watch sports), hold certain attitudes or beliefs (e.g., identify as Republicans), or are customers of a particular company (e.g., buy Dr. Pepper).

A *proprietary panel* is a type of specialty panel in which the members of the panel participate in research for a particular company (e.g., a vehicle manufacturer gathers email addresses of high-end vehicle owners who have volunteered to take surveys about vehicles).

Targeted samples may select panel members who have characteristics of specific interest to a researcher such as auto owners, specific occupational groups, persons suffering specific diseases, or households with children (often this information has been gathered in prior surveys). In addition, regardless of how the sample is selected, individual surveys may include specific screening criteria to further select for low incidence populations.

In the following sections we describe the most commonly-used approaches to building, managing, and maintaining online panels. While Couper (2000) has offered a comprehensive typology of Internet samples, this report focuses primarily on online panels. Other online sample sources such as customer lists or Web options in mixed-mode surveys are not discussed. Because the vast majority of panels do not rely on probability-based methods for recruitment we discuss those first. We then describe the probability-based model. We conclude with a discussion of *river sampling*. Although this method does not involve panel development per se, it has become an increasingly popular technique for developing online samples.

NONPROBABILITY/VOLUNTEER ONLINE PANELS

The nonprobability volunteer online panel concept has its origins in the earlier mail panels developed by a number of market research companies, including Market Facts (now Synovate), NOP, and NFO. These panels generally had the same nonprobability design as contemporary online panels and were recruited in much the same way only relying on offline sources. While many of these panels originally were built to support syndicated research¹, they came to be widely used for custom research² as well. Their advantages for researchers were much the same as those touted for online panels: (1) lower cost; (2) faster response; and (3) the ability to build targeted samples of people who would be low incidence in a general population sample (Blankenship, Breen, and Dutka, 1998).

Today, companies build and manage their online panels in a number of different ways and draw on a wide variety of sources. There is no generally-accepted best method for building a panel and many companies protect the proprietary specifics of their methods with the belief that this gives them a competitive advantage. There are few published sources (see, for example, Miller, 2006; or Comley, 2007) to turn to and so in this section we also rely on a variety of informal sources, including information obtained in RFPs, technical appendices in research reports, and informal conversations with panel company personnel.

Overall we can generalize to five major areas of activity: (1) recruitment of members; (2) joining procedures and profiling; (3) specific study sampling; (4) incentive programs; and (5) panel maintenance.

RECRUITMENT OF MEMBERS. Nonprobability-based panels recruit their members in a variety of ways, but ultimately all involve a voluntary self-selection process on the part of the person wanting to become a member. Panel companies all try to put the invitation to join in front of as many people as possible. The choice of where and how to do this is guided by a combination of cost effectiveness and the desired demographic and behavioral characteristics of recruits. Both online and offline methods may be used. For specialty panel recruitment, this may mean a very selective recruitment campaign (e.g., involving contact with specific organizations or ad buying for specific Web sites, magazines, or TV programs that are targeted to draw people with the characteristics of interest). Regardless of the medium, the recruitment campaign typically appeals to some combination of the following motivations to complete surveys:

1. A contingent incentive, either fixed (money or points received for each completed survey that can be redeemed after a certain number of completed surveys) or variable (sweepstakes with the potential to win prizes for each completed survey);
2. Self-expression (the importance of expressing/registering one's opinions);
3. Fun (the entertainment value of taking surveys);
4. Social comparison (the opportunity to find out what other people think); and
5. Convenience (the ease of joining and participating).

¹ Also sometimes referred to as "multi-client research," this is a market research product that focuses on a specific topic or population but is conducted without a specific client. Instead, the company designs and conducts the research on its own and then sells it to a broad set of clients.

² Research done on a specific population and topic and sponsored by a client.

Although it is widely assumed that earning incentives is the primary motive for joining a panel, there are a few studies to help us understand which motivations are most prominent. Poynter and Comley (2003) report a mix of respondent motives in their study, with incentives topping the list (59 percent) but significant numbers reporting other factors such as curiosity (42 percent), enjoying doing surveys (40 percent), and wanting to have their views heard (28 percent). In a follow-on study, Comely (2005) used results from an online study of panelist motivation to assign respondents to one of four segments: the “Opinionated” (35 percent) who want to have their views heard and enjoy doing surveys; “Professionals” (30 percent) who do lots of surveys and generally will not respond unless there is an incentive; the “Incentivized” (20 percent) who are attracted by incentives but will sometimes respond when there isn’t one; and “Helpers” (15 percent) who enjoy doing surveys and like being part of the online community.

One very popular method for online panel development is through co-registration agreements. Many sites compile email databases of their visitors through a voluntary sign-up process. Portals, e-commerce sites, news sites, special interest sites, and social networks are all examples of sites with a large volume of traffic that the site owner might choose to “monetize” by presenting offers that include those to join a research panel (meaning that the site receives some financial compensation for each person recruited from its site). As a visitor registers with the site, he or she may be offered the opportunity to join other “partner” company databases. The invitation to join additional email lists often asks the user if he or she is interested in receiving special offers or information from partner companies. Panel companies purchase these email address lists from the site owner and contact individuals with offers to join their panel, or users may be taken immediately to the panel company’s registration page. A classic example of this approach is the original panel developed by Harris Black International. This panel was initially recruited through a co-registration agreement with the Excite online portal (Black and Terhanian, 1998).

Another commonly used approach is the use of “affiliate hubs.” These are sites that offer access to a number of different online merchants. A visitor who buys merchandise or services from a listed merchant through the hub receives points that can then be redeemed for merchandise at the hub. (One example typical of such hubs is www.mypoints.com.) Panel companies will sometimes post their offers on these hubs alongside those of various online merchants. Interested visitors can click through from the hub to panel’s registration page. In addition, hubs often conduct email campaigns with registered visitors advertising opportunities to earn points. These might include survey initiations in which potential participants are offered incentives in the site’s currency.

Panel companies may also recruit online via display ads or banners placed across a variety of sites. The panel company generally will not place these ads directly. Rather, the company buys placements from one of the major online advertising companies who in turn place ads where they expect to get the best return, that is, click-throughs to the panel’s registration page.

Still another method relies on search engines. A panel company may buy text ads to appear alongside search engine results with the expectation that some visitors will see the ad and click through to join the panel. These frequently are tied to the use of specific search terms such as “survey” or “market research.” Search for “survey research” on search engines like Yahoo, Google, or Bing and you likely will see at least one offer on to join a panel on the advertisement section of the search results page.

Though not used by any reputable company in the U.S., other email recruitment tactics include blasting or spamming methods – sending unsolicited commercial email in mass quantities. Sending these

requests en masse creates the potential for violation of the CAN-SPAM Act of 2003 and may result in ISP blacklisting, which blocks the sender's email systems from sending out any further emails.

Finally, as has been often done in research in other modes under the names of snowball recruiting or viral recruiting, some panel companies encourage their members to recruit friends and relatives. These programs often offer the member a per-recruit reward for each new member recruited.

There is an equally wide variety of methods for offline recruitment. Many full-service research companies conduct surveys in a variety of modes (including paper-pencil and RDD telephone research as well as online surveys). Sometimes they will field a survey that has as its primary purpose the recruitment of panel members (known as *purposeful recruitment*). Other times, they may field a study that will recruit panel members as a byproduct of their traditional research activities (known as *incidental recruitment*). For example, they might routinely ask a respondent at the end of a RDD telephone survey whether he or she is interested in joining a panel to take future surveys online. Or, a company might borrow a common technique from telemarketers and use an IVR system to autodial numbers and invite people to join a panel. In the event the contacted person is interested, the system asks for a name and email address for future contact that can then be used to send an email invitation to join the panel.

Companies also may include invitations to join a panel in offline marketing or advertising campaigns, such as messaging on cash register receipts, event tickets, monthly account statements, and various other traditional media. Sometimes surveys are conducted for a specific company (with invitations to the survey on cash register receipts), and upon completion of the survey, respondents are offered the opportunity to join a panel (whether general population or specialty panel). Merchants who allow panel companies to do this often get a discount on their research costs for each recruited respondent. Yet another source for panel members occurs with some companies that are so large and feature such diverse products that they convert their databases into panels and develop a supplemental line of business.

No two panels recruit their members with the same mix of techniques. Individual panels set targets for the specific populations and target respondents they feel will represent the most compelling value proposition to research clients and design their recruitment strategies appropriately. However, the variation in recruitment sources and methods by the various companies can lead to differences in research findings, something we address in Section 4 on errors of nonobservation.

Panel companies rarely disclose the success rates from their recruitment strategies. One exception is a study by Alvarez, , and Van Beselaere (2003) based on a project conducted by a team of academics who built an online panel for the Internet Surveys of American Opinion project. Through a co-registration agreement with ValueClick, Inc., an online marketing company with access to a wide variety of Web sites, Internet users who were registering for various services were provided a check box on their registration form to indicate their interest in participating in Web surveys. Some 21,378 Web users (out of an unknown total number of people passing through the targeted registration sites) checked the box. Among this group, 6,789 completed the follow-up profile and were enrolled in the panel (a 32 percent yield).

Alvarez et al. also collected data on the effectiveness of banner ads. Their banner ad was displayed over 17 million times, resulting in 53,285 clicks directing respondents to the panel Web site, and ultimately 3,431 panel members. The percentage yield of panel members per click was 6.4 percent, and the percentage member yield per banner display (a.k.a. impression) was 0.02 percent.

JOINING PROCEDURES AND PROFILING. Joining a panel is typically a two-step process. At a minimum, most reputable research companies in the U.S. follow what is called a *double opt-in* process whereby a person first indicates his/her interest in joining the panel (either signing up or checking a box on a co-registration site). The panel company then sends an email to the listed address and the person must take a positive action indicating the intent to join the panel. At this second stage of confirmation some panel companies will ask the new member to complete a profiling survey that collects a wide variety of background, demographic, psychographic, attitudinal, experiential, and behavioral data that can be used later to select panelist for specific studies. This double opt-in process is required by ISO 26362 (International Organization for Standardization, 2009) and the industry has come to accept it as defining the difference between a panel and simply a database of email addresses. This international standard was released in 2009 as a supplement to the previously-released ISO 20252 – Market, Opinion, and Social Research (International Organization for Standardization, 2006). ISO 26362 specifies a vocabulary and set of service quality standards for online panels.

Upon agreeing to join, panelists typically are assigned a unique identification number used to track the panelist throughout their lifetime on the panel. Panel companies then assign respondents to their correct geography and demographic groups so that they can provide samples by DMA, MSA, ZIP code, and other geographic identifiers.

As part of this initial recruitment most panel companies now have validation procedures to ensure that individuals are who they say they are and are allowed to join the panel only once. Checks at the joining stage may include: verification against third party databases, e-mail address validity (via format checks and checks against known ISPs), postal address validity (via checks against postal records), “reasonableness” tests done via data mining (appropriate age compared to age of children, income compared to profession, etc.), duplication checks, or digital fingerprint checks to prevent duplication by IP address and ensure correct geographic identification.

Panel companies vary in the type of information they collect in their respondent profiles. A company may try to create a panel database with hundreds of variables so that they can select efficient, targeted samples. Profiles might include shopping habits, product brand ownership, health ailments, occupation, and household roles, to name a few. Companies then price their services based on the incidence of the specific groups of interest to sample buyers. Panel companies typically refresh these profiles on a regular basis, sometimes simply to update and other times to add still more information about panelists. They also may capture screening information from individual surveys and incorporate those data into member profiles.

SPECIFIC STUDY SAMPLING. Simple random samples from panels are rare because of the tendency for them to be highly skewed toward certain demographic characteristics (e.g., just as older people are more likely to answer landline phones, younger people are more likely to be online and respond to an email survey invitation). *Purposive sampling* (discussed in more detail in Section 6) is the norm and the sampling specifications used to develop these samples can be very detailed and complex, including not just demographic characteristics but also specific behaviors or even previously-expressed positions for specific attitudes.

Although the empirical evidence is mixed, some researchers believe that having respondents take too many surveys in too short of a time interval can yield results that are less than optimal (Garland, Santus, and Uppal, 2009). To prevent some respondents from “over-participating” and dominating results, some

panel suppliers include so-called “lockout” criteria that prevent some panel members from being sampled. These lockouts might specify that no one who has completed a survey on the same topic during a specific time frame can be sampled for the current study (e.g. cannot participate in another political survey for three months). Or the panel company may limit the frequency with which a member can be solicited for a survey (e.g., they will not invite any member to a new survey for at least 10 days).

With the sample drawn, the panel company sends an email invitation to the sampled member. The content of these emails varies widely and generally reflects the information the panel company’s client wishes to put in front of the respondent to encourage participation. At a minimum the email will include the link to the survey and a description of the incentive. It might also specify a closing date for the survey, the name of the organization conducting or sponsoring the survey, an estimate of the survey length, and even a description of the survey topic.

INCENTIVE PROGRAMS. To combat panel attrition, and to increase the likelihood that panel members will complete a survey, panelists are typically offered compensation of some form. These include cash, points redeemed for various goods (e.g., music downloads, airline miles, etc.), sweepstakes drawings, or instant win games. Incentive programs vary from company to company but the amount of incentive is typically tied to the length or topic of the survey. Large incentives may be used when the expected incidence of qualifiers is low. These incentives are paid contingent on completion, although some panels also pay out partial incentives when a member starts a survey but fails to qualify.

PANEL MAINTENANCE. Once people have been recruited to join a panel, the challenge is to keep them active. People who join expect to take surveys. Each panel company has its own definition as to what it deems as an active panelist. However, nearly all of the definitions are based on a calculation that balances the date a person joined and the number of surveys taken in a specified time period. ISO 26362 defines an active member as a panelist who either has participated in at least one survey or updated his/her profile in the last year. As such, the claims of panel size depend greatly on value of the variables in the algorithm used to calculate the status of a member.

Most panel companies have a multi-faceted approach to maintain a clean panel comprised of members whose information is current and who can be contacted successfully to take a survey. These “hygiene” procedures include treatment plans for undeliverable e-mail addresses, “mailbox full” status, syntactically undeliverable email addresses, non-responding panelists, panelists with missing data, panelists who repeatedly provide bad data, and duplicate panelists. In addition, panel companies may expend considerable effort to maintain deliverability of panelist email addresses via white-listing³ with major ISPs.

Attrition of panel members is a natural process with the levels varying considerably from one panel to the next. As occurs with most joining activities, the greatest attrition most likely occurs among the newest members. Attrition can come from multiple sources – people change email addresses but do not update with the panel company, they can drop out due to long or boring surveys, they switch panels because they may have better rewards or more interesting or shorter surveys, etc. Once a panel has been established, panel recruitment becomes an essential ongoing activity to ensure a sufficient number of respondents are available for ongoing projects.

³ An email white list is a list of email addresses from which an individual or an ISP will accept email messages.

In light of attrition and underutilized panelists, firms attempt to reengage people who appear to be reluctant to participate in surveys and those who have left the panel altogether. But there appears to be nothing in the research literature reporting the effectiveness of such approaches.

PROBABILITY-BASED RECRUITMENT

Despite the early emergence of the Dutch Telepanel in 1986 (Saris 1998), online panels that recruit using traditional probability-based methods have been slow to appear and are fewer in number than volunteer panels, although they are now gaining in prevalence. These panels follow roughly the same process as that described for volunteer panels with the exception that the initial contact with a potential member is based on a probability design such as RDD or area probability. To account for the fact that not everyone in the sample may have Internet access, some of these panels provide the necessary computer hardware and Internet access or may conduct surveys with the panels using a mix of modes (Web, telephone, mail, IVR, etc.). In one study of the four stages of the recruitment process (recruitment, joining, study selection, participation), Hoogendoorn and Daalmans (2009) report that there is differential non-response and engagement at each stage for a probability panel associated with a number of demographic variables, including age and income. The effects of this self-selection on population estimates for survey measures in subsequent surveys remain unexplored.

Aside from the key differences in sampling and provision of Internet access to those who are not already online, probability-based panels are built and maintained in much the same way as nonprobability panels (Callegaro and DiSogra, 2008). Once contacted, a respondent can indicate his/her willingness to join the panel in a number of ways – by agreeing in the telephone interview, calling a toll-free number, completing and returning the mail solicitation, or going to the registration page of the panel's Web site. Once registered, the panelist is asked to complete an online profile questionnaire much like those used by panel companies generally. Sampling methods and incentive structure vary depending on individual study requirements. As with any panel, attrition creates the need for ongoing recruitment and rebalancing. In addition, because the cost of acquisition of panel members is higher than it is for nonprobability panels, probability panels may require completion of a minimum number of surveys on a regular basis to remain in the panel.

The combination of the cost of recruitment and the requirement of some panels to provide Internet access for those who are not already online generally translates to these panels being more expensive to build and maintain than those that use nonprobability methods. As a consequence, their members may number in the tens of thousands rather than the millions often claimed by volunteer panels. Studies executed with these panels tend toward smaller sample sizes, although, depending on the target respondent, studies with large sample sizes are possible. It also can be difficult to get large sample sizes for low-incidence populations or smaller geographic areas unless the panel was designed with these criteria in mind. Nonetheless, probability-based panels are attractive to researchers who require general population samples and a basis in probability theory to ensure their representativeness.

RIVER SAMPLING

River sampling is an online sampling method that recruits respondents when they are online and may or may not involve panel construction. Sometimes referred to as *intercept interviewing* or *real-time sampling*, river sampling most often will present a survey invitation to a site visitor while he/she is engaged in some other online activity.⁴ In the vast majority of river sampling applications, the target population is much broader than visitors to a single site and determining how many and on which Web sites to place the invitation is a complex task. Knowledge about each site's audience and the response patterns of their visitors is a key piece of information needed for effective recruiting. Companies that do river sampling seldom have access to the full range of sites they need or the detailed demographic information on those sites' visitors, and so they work through intermediaries. Companies such as Digitas and DoubleClick serve up advertising across the Internet and also will serve up survey invitations. ISPs such as AOL or Comcast might also be used.

Once the sites are selected, promotions and messages of various types are randomly placed within those sites. The invitation may appear in many forms, including within a page via a randomized banner, an *n*-th user pop-up, or pop-under page. In many cases, these messages are designed to recruit respondents for a number of active surveys rather than a single survey. Visitors who click through are asked to complete a short profile survey that collects basic demographics and any behavioral data that may be needed to qualify the respondent for one or more of the active surveys. Once qualified, the respondent is assigned to one of the open surveys via a process sometimes referred to as *routing*.

Upon completion of the survey respondents are generally, but not always, rewarded for their participation. River sampling incentives are the same as those offered to online panel members. They include cash, PayPal reimbursements, online merchant gift codes and redeemable points, frequent flyer miles, and deposits to credit card accounts.

There are some indications that river sampling may be on the rise as researchers seek larger and more diverse sample pools and less-frequently surveyed respondents than those provided by online panels.

⁴ Many Web sites use pop-up questionnaires to survey visitors to ask them about their own Web site (e.g., are the features easy to access, did they obtain the information they were looking for, etc.) and these surveys are known as *Web site evaluations*. Distinct from Web site evaluations, the invitations to a river sample direct the visitor away from the originating Web site to a survey that is not about the Web site or Web site company.